IMD3002 - 3D COMPUTER GRAPHICS

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L21 – Rendering: Advanced Techniques

Overview

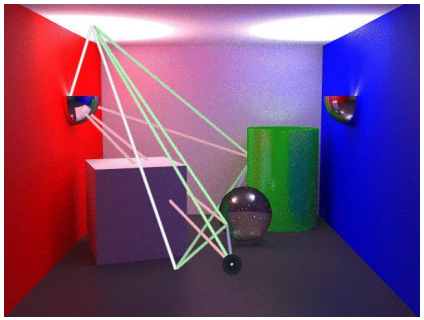
- This lecture introduces a variety of other rendering methods including: Path Tracing, Ambient Occlusion, Final Gather, Caustics, and Image-Based Rendering

Path Tracing

- Introduction
  - Path Tracing is similar to Ray Tracing
  - Rays are cast from a camera and traced through a 3D scene
  - Path tracing simulates many elements that are required to be specifically added to by methods (such as traditional ray tracing or scan-line rendering)
    - These include: depth of field, soft shadows, caustics motion blur, and indirect lighting
  - Path Tracing includes these effects making rendering much simpler

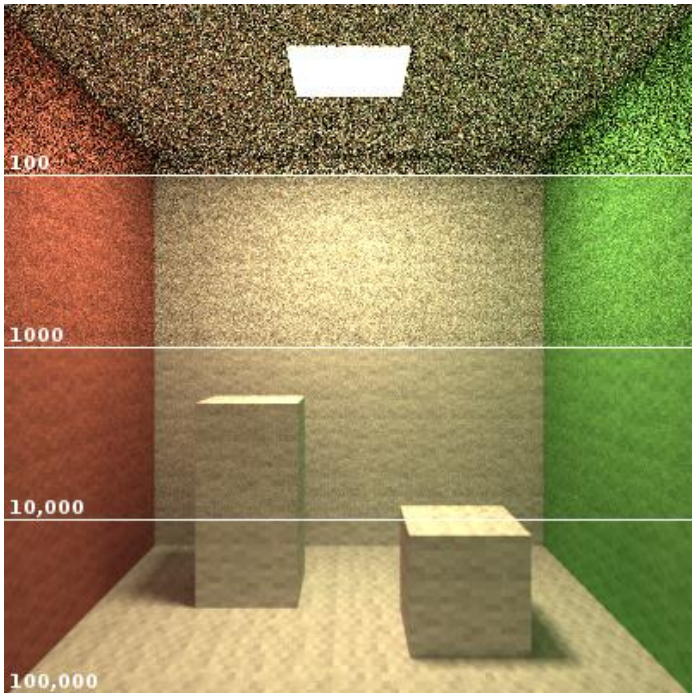
Path Tracing

- Process
  - Path Tracing casts rays from a virtual camera and traced them through a virtual scene
    - The reflected ray distribution is randomly emitted from each source pixel and at each intersection with an polygon in the scene
    - A ray eventually exits the scene or terminates
    - Once the ray has finished bouncing around a sample value is calculated from the materials the ray interacted with against
    - The sample value is added back to the average for the source pixel



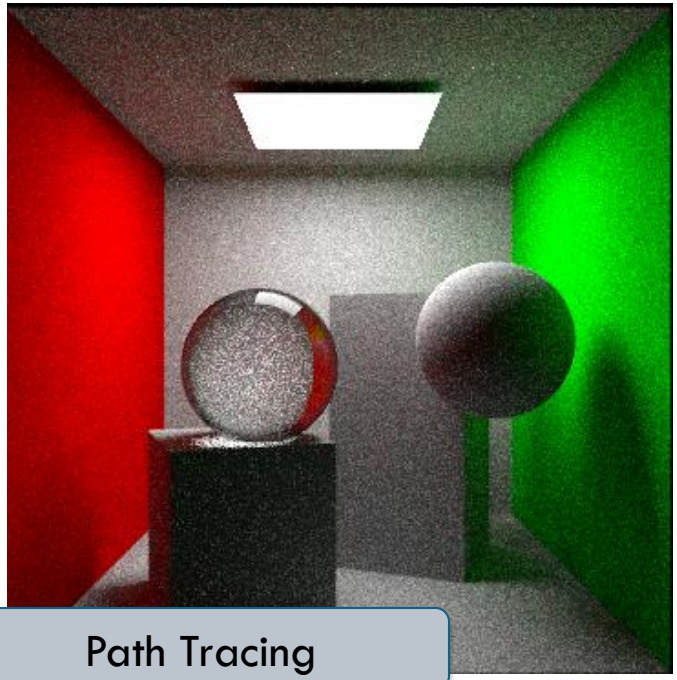
Path Tracing

- Other
  - The random element in path tracing causes the rendering to be noisy
  - Noise decreases over as sampling increases



Path Tracing

- Bidirectional Path Tracing
  - Bidirectional Path Tracing (BDPT) can handle indirect lighting problems far more robustly and efficiently compared to ordinary path tracing



Path Tracing

Bidirectional Path Tracing

Path Tracing

- Metropolis Light Transport
  - The procedure constructs paths from camera to light using BDPT, with slightly modified paths
  - The Metropolis algorithm (a statistical algorithm) is used to compute the appropriate distribution of brightness over the image
  - It has the advantage over bidirectional path tracing, in that once a path has been determine from light to camera, it can then explore nearby paths; thus obscure light paths can be more thoroughly determined using same number of simulated photons

Path Tracing

- Metropolis Light Transport
  - The algorithm essentially generates a path and stores the path's 'nodes' in a list; it can then modify the path by adding extra nodes and creating a new light path
  - By creating this new path, the algorithm decides how many new 'nodes' to add and whether or not these new nodes will actually create a new path
  - It is an unbiased method that, in most cases, converges to a solution of the rendering equation faster than other unbiased algorithms such PT or BDPT

Path Tracing

- Use in Industry
  - Many companies in the movie industry (and possibly soon the game industry) are using path tracers over ray tracing
  - Pixar/Disney have also started to use Path Tracing and have reworked **Renderman** to include this method, it was first used on Monster's University
  - The **Arnold** Renderer is a good example of a GI system using Path Tracing

Final Gather

- Introduction
  - Final Gather (FG) is an additional tool for indirect illumination
  - FG calculates the diffuse irradiance from indirect light in the scene without considering the caustic effects on glossy to specular surfaces
  - FG does not rely on photons to calculate the illumination, instead it uses ray tracing to trace rays from a surface outward into the scene

Final Gather

- Introduction
  - FG does not solely depend on light sources for illumination; it reflects light from various sources that either are affected by a light source or that generate their own radiance by means of a shader
  - The diffuse and specular colour components of a shader require a light source in order to appear visible when rendered
  - FG lighting does not produce specular highlights

Final Gather

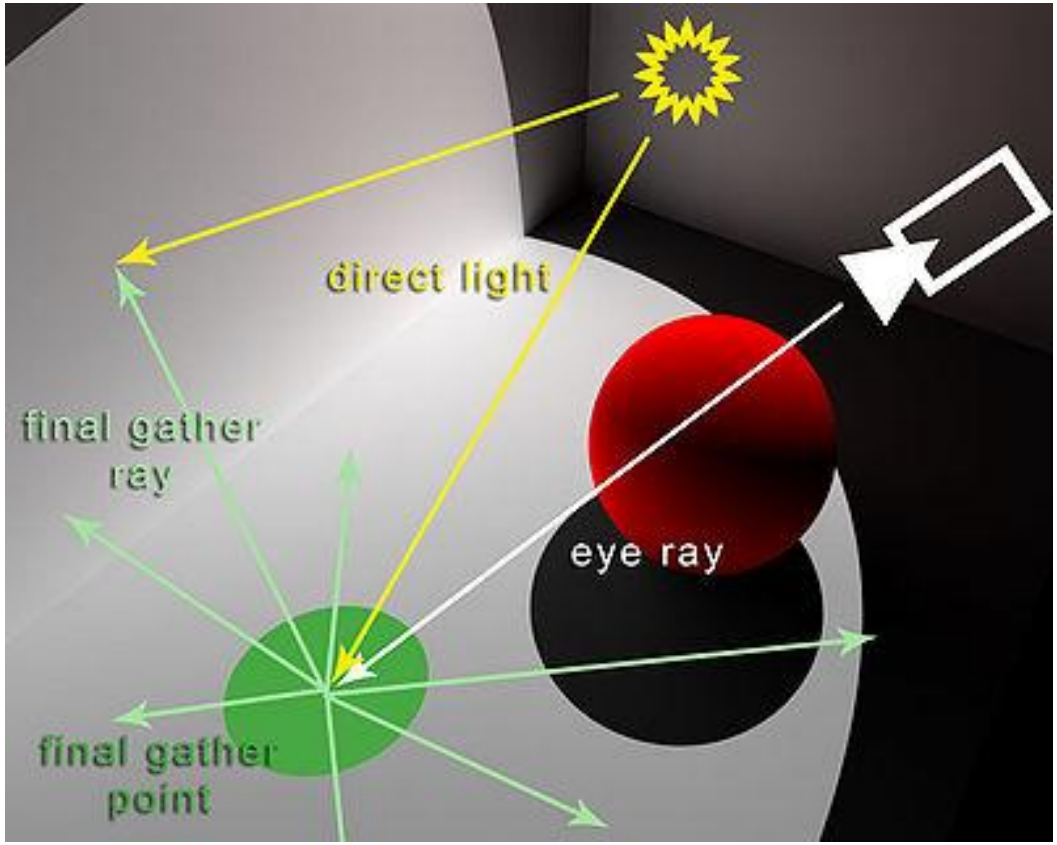
- Elements used for Generation
  - The following elements are used by FG to generate indirect diffuse lighting:
    - Surfaces lit by light sources
    - Surfaces that reflect ambient light
    - Environment images, or constant background shading
    - Surfaces that have constant shading and light emitting surfaces

Final Gather

- Process
  - The purpose of FG is to provide a <u>final</u> pass that <u>gathers</u> irradiance values in the scene prior to rendering
    - This is why it is called "Final Gather"
  - Final Gather is a two-stage process:
    - **Pre-processing FG points** in the scene (distributing FG points and gathering irradiance)
    - **Interpolating the irradiance** (both colour and brightness) values from FG points during the rendering phase

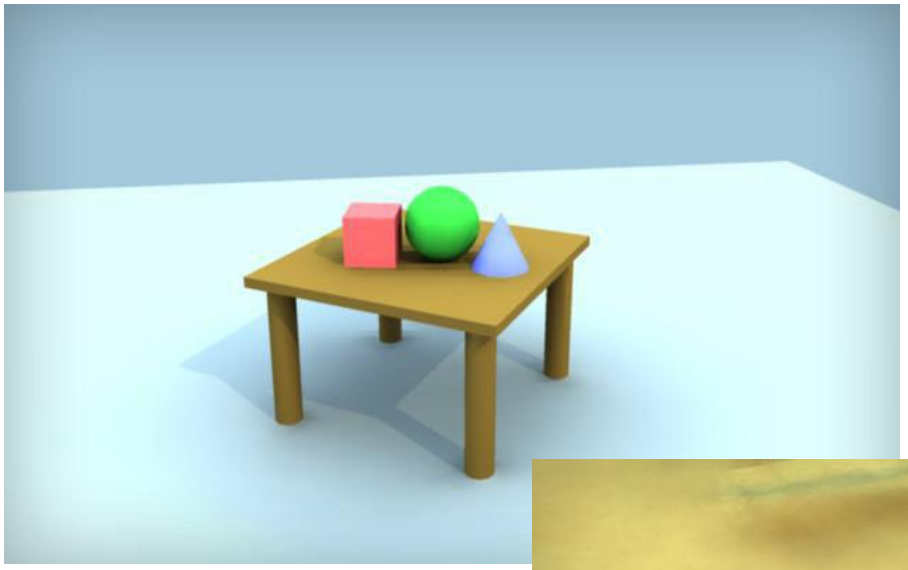
Final Gather

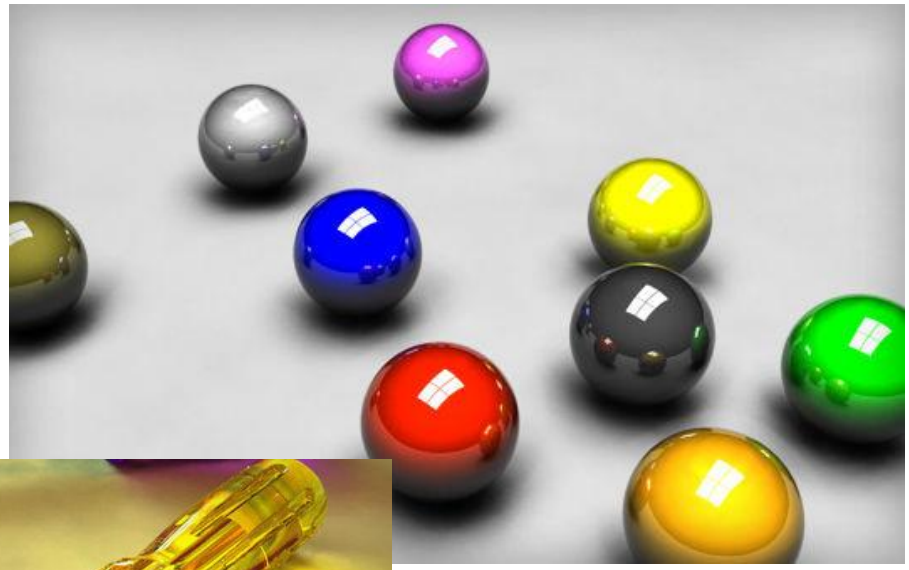
- Process
  - FG uses (backwards) ray tracing instead of photons



Final Gather

- Results





Irradiance Particles & Importons

- Introduction
  - Irradiance particles is a relatively new method for global illumination based on importance sampling
  - It is generally easier to set up and faster than other methods

Irradiance Off

Irradiance Single Pass

Irradiance 2 Rays

Irradiance 512 Rays

Irradiance Particles & Importons

- Importons
  - Importons are particles similar to photons
  - However, they are traced from the camera and traverse the scene in the opposite order
  - Instead of energy, they hold a quality which is an importance of the contribution to the final image
  - The information from importons is used by the rendering system based on effort according to the final contribution to the rendered image, in other words, importance-driven sampling

Irradiance Particles & Importons

- Importons
  - In the context of irradiance particles, importons can be considered as locations that are used to store indirect light
  - It is important that all parts of the scene are covered by importons, particularly those parts seen by the camera, but indirectly as well (e.g. by a mirror)

Irradiance Particles & Importons

- Process
  - Irradiance particles is performed in multiple preprocessing steps:
    - Importons are emitted from the camera to the scene, which are collected as irradiance particles
    - The irradiance particles are set into a “particle map” containing information about direct illumination at that position
    - Depending on how many light bounces you choose to include, several other passes are processed to also include the indirect irradiance of these particles
    - If interpolation is used, a final preprocessing step is done in which irradiance is collected for every irradiance particle in order to be used later during rendering

Irradiance Particles & Importons

- Process
  - After this preprocessing is done, the particles may be saved in a file
  - During rendering, the particle map is used to estimate the indirect illumination (the irradiance) for every shading point

Irradiance Particles & Importons

- Settings/Controls
  - The following elements are controllable for importons:
    - **Density** ~ number of importons shot from the camera per pixel (nominally far less than 1, e.g. 0.04)
    - **Merge Distance** ~ the distance at which the particles are merged
    - **Max Depth** ~ sets the amount of importon diffusion (the default value of zero means no scattering on difusse bounces)

Caustics

- Introduction
  - Caustics are an addition to GI rendering (e.g. ray tracing) that calculates photons of light starting at a light source (like real light), which can be reflected, refracted, bounced off mirrors, or concentrated by a lens, accurately simulating more of the ways real light can move through a scene
  - One of the most common methods for generating caustics is the use of photon mapping, which we have discussed in the previous lecture

Caustics

- Introduction
  - **Example:** the rendering on the next page combines ray-tracing (for the mirrors, shadows, refractive vase) and photon mapped caustics (for the light reflected off the mirror onto the side wall, the light focused by refraction through the vase, and the light reflected onto the top of the dresser by the vase)

Caustics

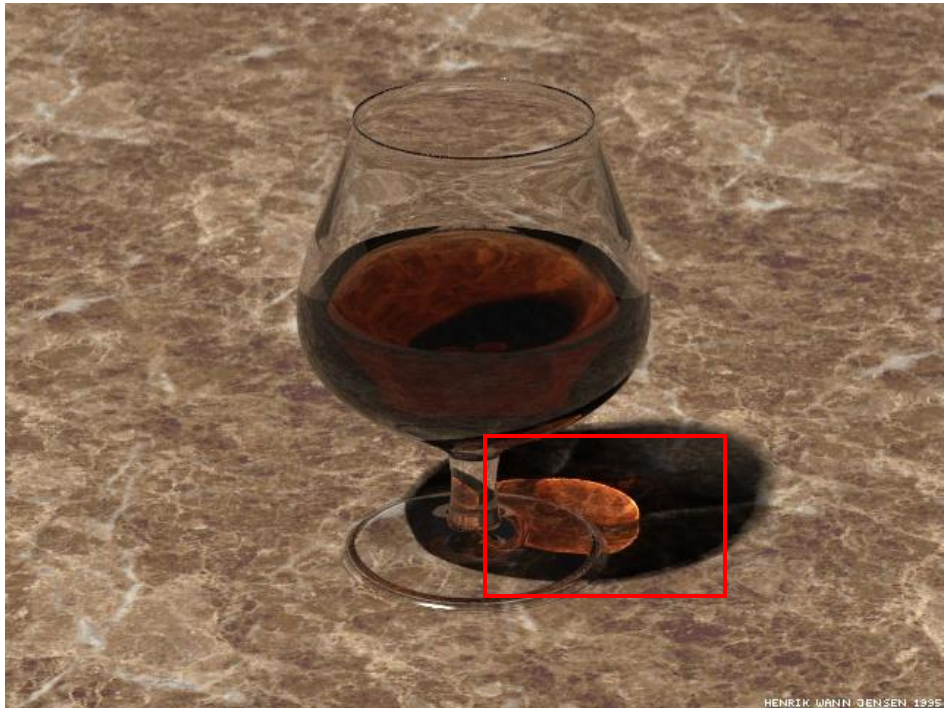
- Introduction

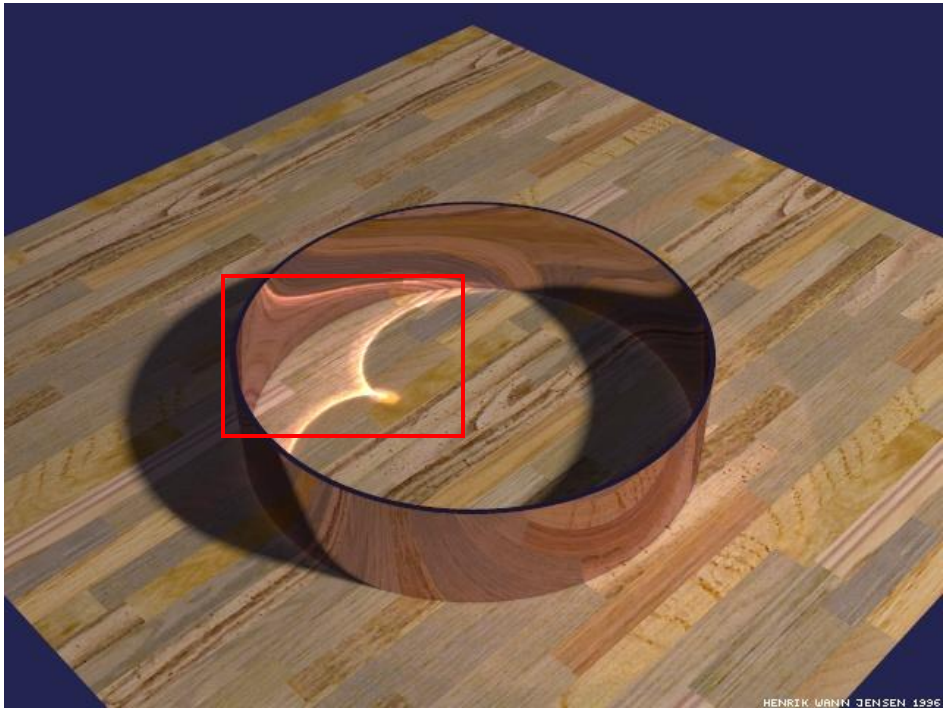
Caustics

- Introduction
  - Here is a comparison between two images, one using only ray-tracing (left), the other with caustics added (the previous image)

Caustics

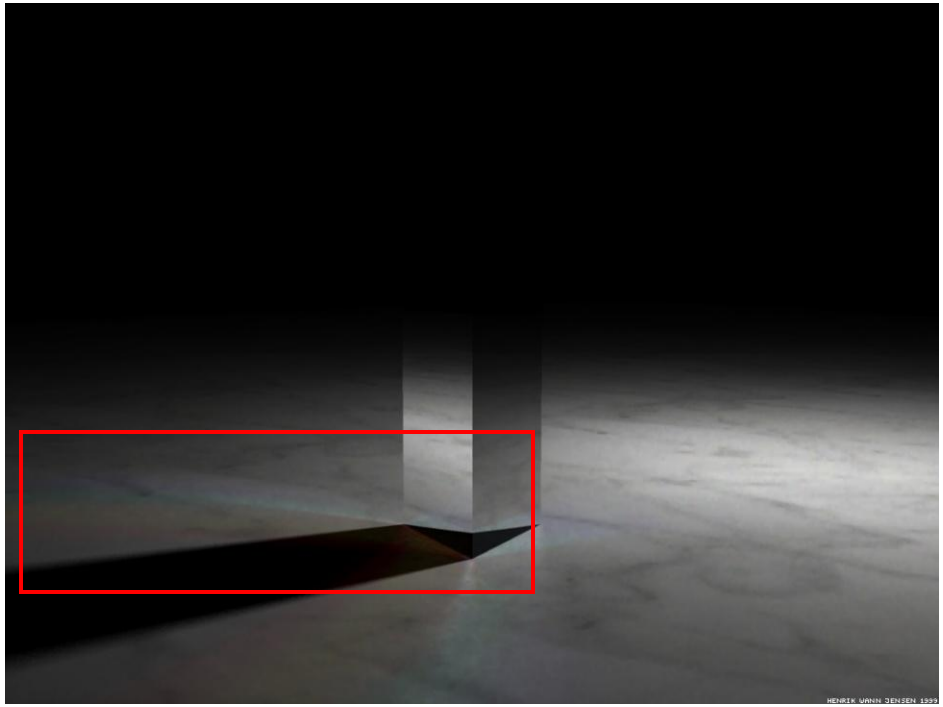
- Introduction
  - **Examples:** a cognac glass, and a copper band; the areas marked would not be seen in regular ray tracing

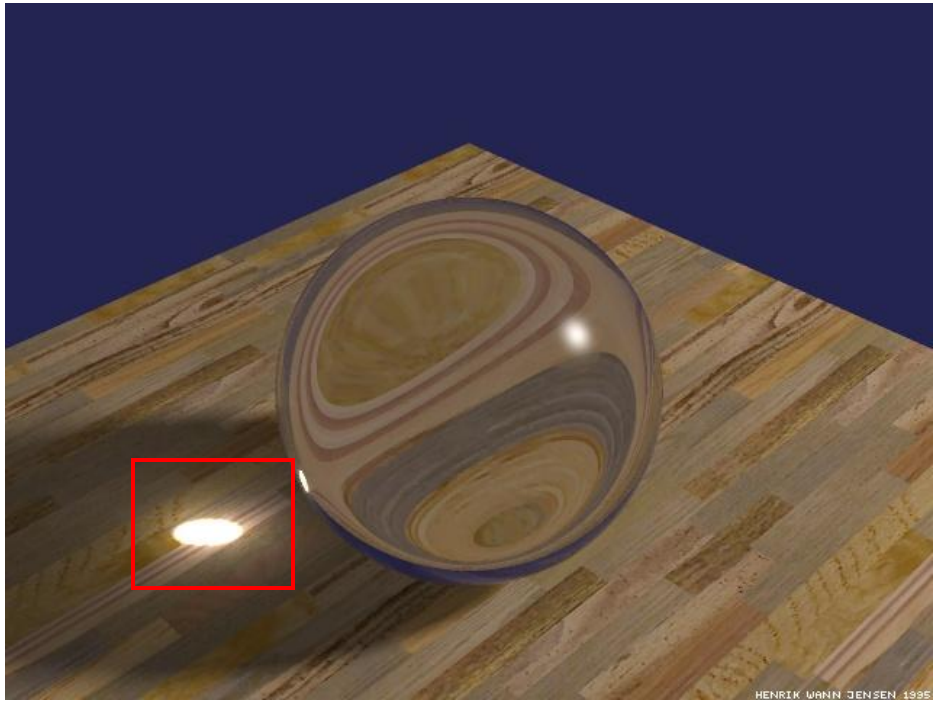




Caustics

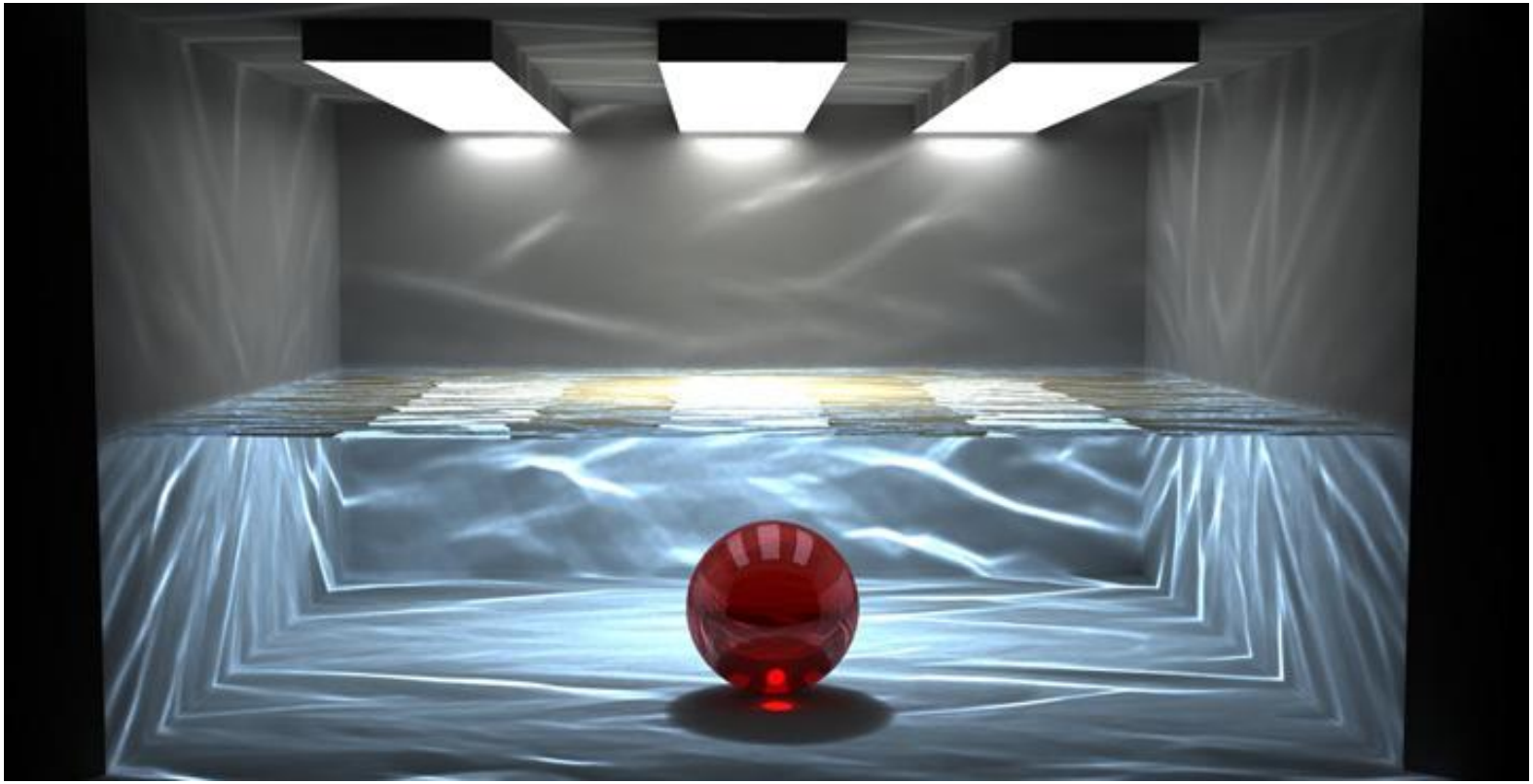
- Introduction
  - **Examples:** prism (with the colour differentiation), and a glass ball (with the light concentration)





Caustics

- Introduction
  - **Example:** water created in 3DS Max and rendered using Mental Ray



Ambient Occlusion

- Introduction
  - Since standard GI models do not take into account diffuse reflection, lighting tends to be darker than real-world lighting
  - Therefore, it becomes necessary to pump up the illumination, and the easiest way to do this is to use the ambient term in the lighting calculations
  - The ambient term adds a uniform intensity value and colour to the surface, ensuring no point on the object is black

Ambient Occlusion

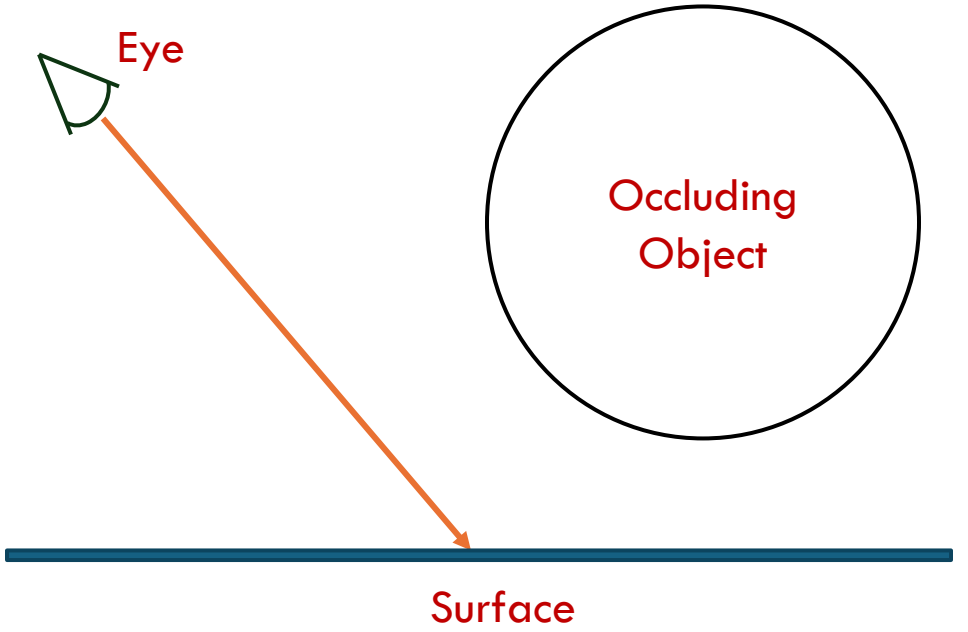
- Introduction
  - Adding a uniform value tends to make areas look flat, and does not take into account cracks, contact and self shadows
  - Therefore, the common technique is to replace ambient light with a lot more diffuse lights
  - This tends to increase the overall render time, which is obviously undesirable
  - To counter this a simple, but highly effective, method called "**Ambient Occlusion**" is used to achieve grounded realism, without the cost of full GI

Ambient Occlusion

- Method
  - Ambient occlusion is simply a ratio of the amount of ambient light a surface point would likely receive
  - It simulates a huge dome light surrounding the entire scene
  - If a surface point is under a foot or table, it will end up much darker than the top of someone's head or the tabletop
  - This can then be multiplied by various other surface attributes to achieve a subtle, but very powerful, lighting effect

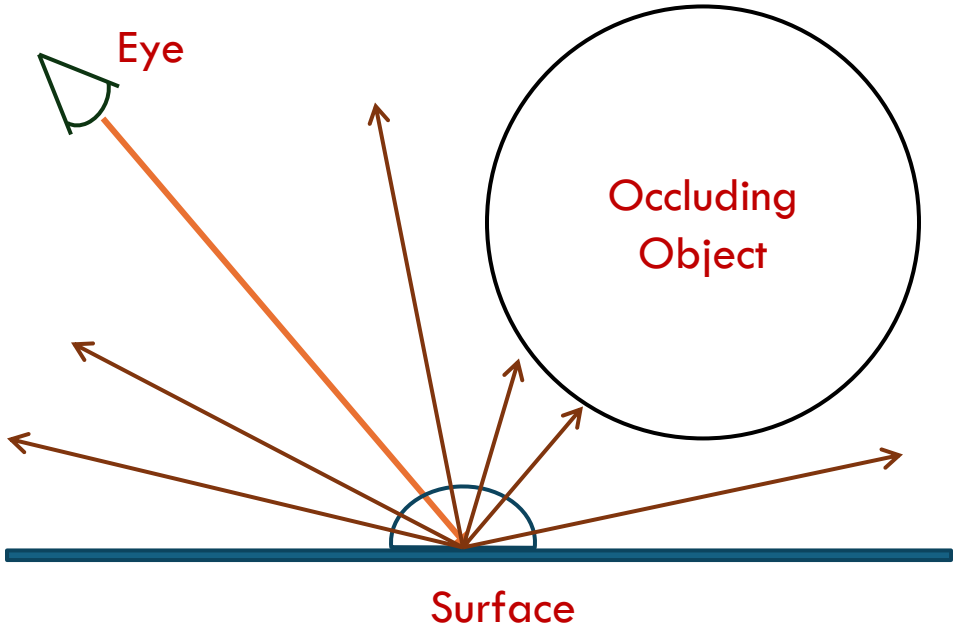
Ambient Occlusion

- Method
  - The initial eye ray intersects with a surface that has an occlusion shader



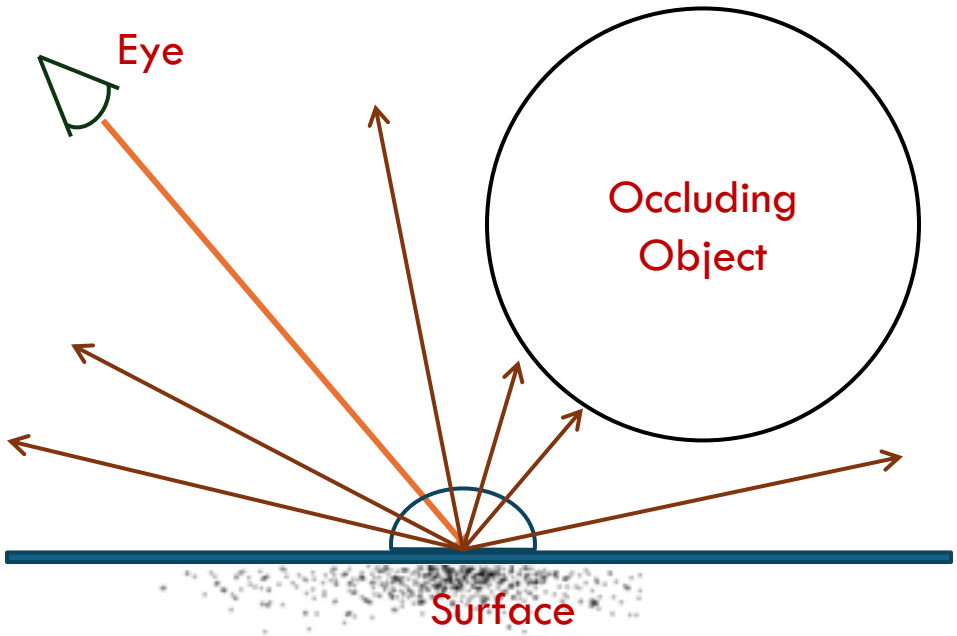
Ambient Occlusion

- Method
  - The surface point sends out a randomised hemispherical spray of rays, biased towards the normal, out to do hit-tests on the rest of the scene



Ambient Occlusion

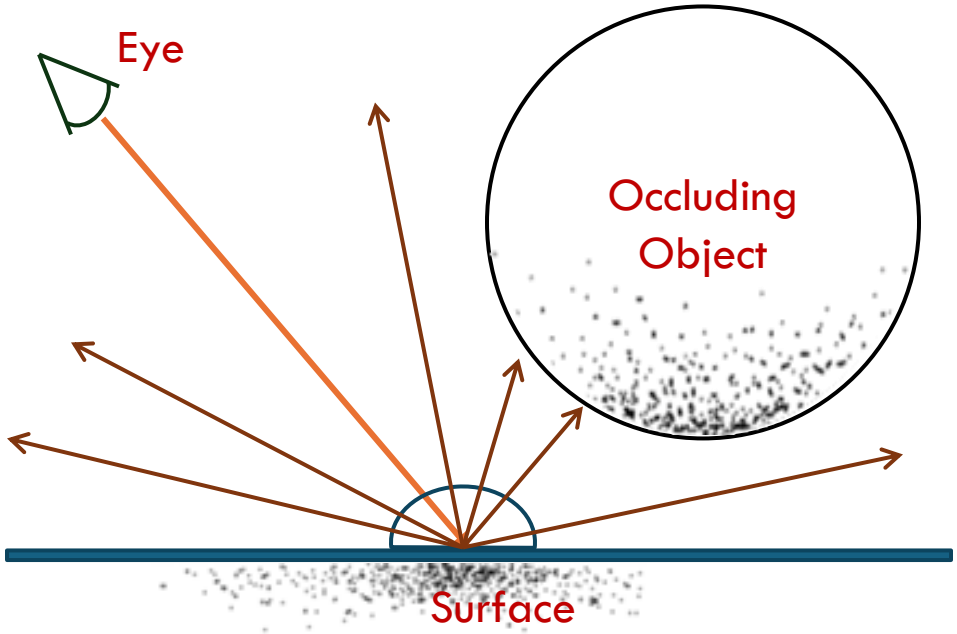
- Method
  - Each surface point is shaded by a ratio of ray intersections to number of original samples



Example: there are 2 intersections for 6 sample rays, creating a ratio of 1/3; subtracting this value from 1.0 an ambient term value for the area in the occluded portions of the surface

Ambient Occlusion

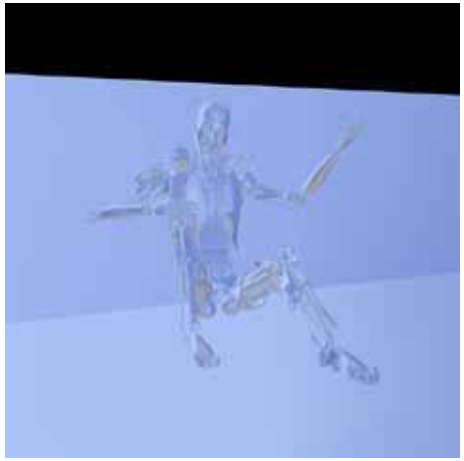
- Method
  - This is then repeated for each object in the scene



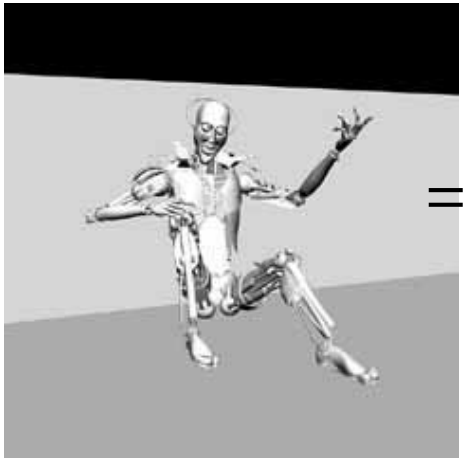
Ambient Occlusion

- Example
  - An ambient occlusion pass, with two additional passes (using Pixar's RenderMan)

Ambient Occlusion



Environment Map



Diffuse Reflection

=

Final Composite

Ambient Occlusion

- Finally …
  - Although Ambient Occlusion was only introduced in 2002 by Industrial Light and Magic (ILM) it is already available in several GI systems and has made a big impact on quick rendering

Image-Based Rendering

- Introduction
  - Image-based rendering (IBR) refers to techniques used to generate new images from other images rather than from geometric primitives
  - These methods combine the provided images and use them to form new images, from different camera locations
  - A closely related concept is image-based modelling (IBM), which has been used to denote any sampled representation of a 3D scene

Image-Based Rendering

- Introduction
  - Current image-based rendering techniques all share several advantages over geometry-based techniques, making them useful for a wide range of situations
  - Image-based techniques are independent of the complexity of the scene to be rendered, making real-time realistic interactions with complex scenes possible

Image-Based Rendering

- Introduction
  - In addition, these methods can make use of rendered images, or images of actual physical environments
  - Therefore, complex and realistic real-life scenes can be viewed without requiring difficult or impossible geometric modelling, and independent of scene complexity

Image-Based Rendering

- Introduction
  - However, current image-based techniques rely on assorted types of sampling and interpolation of source images
  - Therefore, where the input images are too sparse or when they insufficiently sample the environment, errors may occur
  - The image-based techniques that use a small number of input images may fail to accurately capture high frequency components and changes in the scene, such as specular highlights

Image-Based Rendering

- Introduction
  - In general image-based rendering is a little bit outside the scope of this course, it relies more on computer image processing for geometry, but it does have some interesting derivations that we can use: billboarding

QuickTime VR

3D Graphics

Image-Based Rendering

- Billboarding
  - Many special effects, such as lens flare, can be created by rendering an image onto a polygon facing the view
  - Orienting the polygon based on the view direction is called "**Billboarding**", where the polygon is called a billboard
  - As the view changes the orientation of the polygon changes

Image-Based Rendering

- Billboarding
  - Billboarding, combined with alpha texturing and animation can be used to represent many phenomena that do not have a solid surface:
    - Smoke
    - Fire
    - Fog
    - Explosions
    - Energy Shields
    - Vapour Trails
    - Clouds

Image-Based Rendering

- Billboarding
  - In general, most billboarding techniques we determine the surface normal and the up direction
  - These are used to orient the polygon; these two vectors are sufficient to determine the orthonormal basis for the surface; here we shall briefly discuss:
    - Screen-Aligned Billboarding
    - World-Oriented Billboarding
    - Axial Billboard
    - Imposters

Image-Based Rendering

- Billboarding : Screen Aligned
  - The simplest form of billboarding is the use of screen aligned billboards
  - This form is similar to a 2D sprite, in that the image is always parallel to the screen and has a constant up vector (the cameras)
  - Here we determine the surface normal we require as being the negation of the viewing planes normal vector
  - Screen-aligned billboards are useful for information such as annotation text

Image-Based Rendering

- Billboarding : World Oriented
  - Screen aligned billboarding can be used for circular objects, such as particles, as the up vector is irrelevant
  - If the camera rolls, due to the symmetry, the appearance of the billboard doesn't change
  - However, if our shape is not spherical, we need to consider another method which is similar to the screen aligned: world oriented
  - Here we follow the same method, except the up vector is taken from the world, not the camera

Image-Based Rendering

- Billboarding : World Oriented
  - **Example:** clouds created using world-oriented billboards

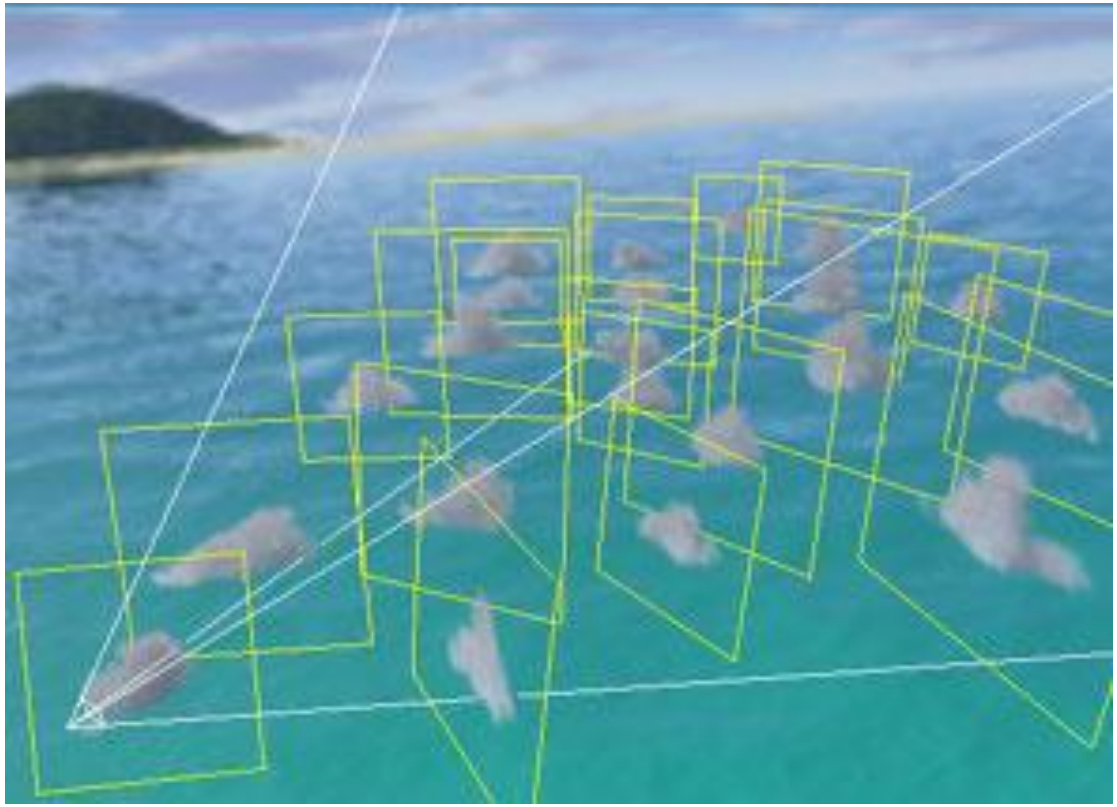


Image-Based Rendering

- Billboarding : Axial Billboard
  - In this scheme the textured object does not normally face straight-on toward the viewer
  - Instead, it is allowed to rotate around some fixed world space axis and align itself so as to face the viewer as much as possible within this range
  - This billboarding technique is commonly used for displaying trees

Image-Based Rendering

- Billboarding : Axial Billboard
  - Instead of representing a tree with a solid surface, or even tree outlines – a single tree billboard is used, with the up vector set to the axis of the tree trunk
    - **Example:** an axial billboard using trees

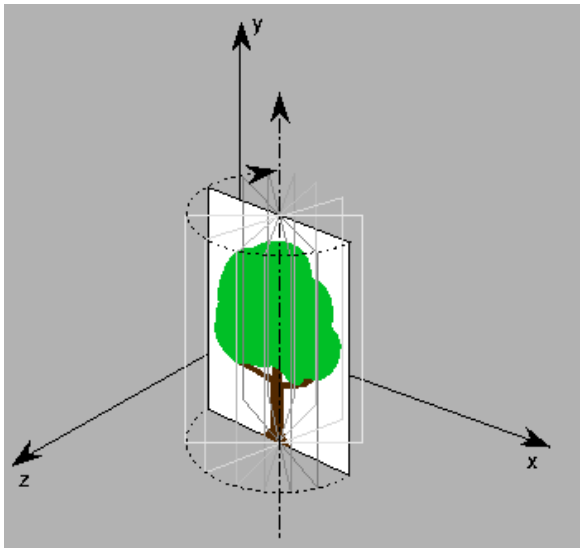


Image-Based Rendering

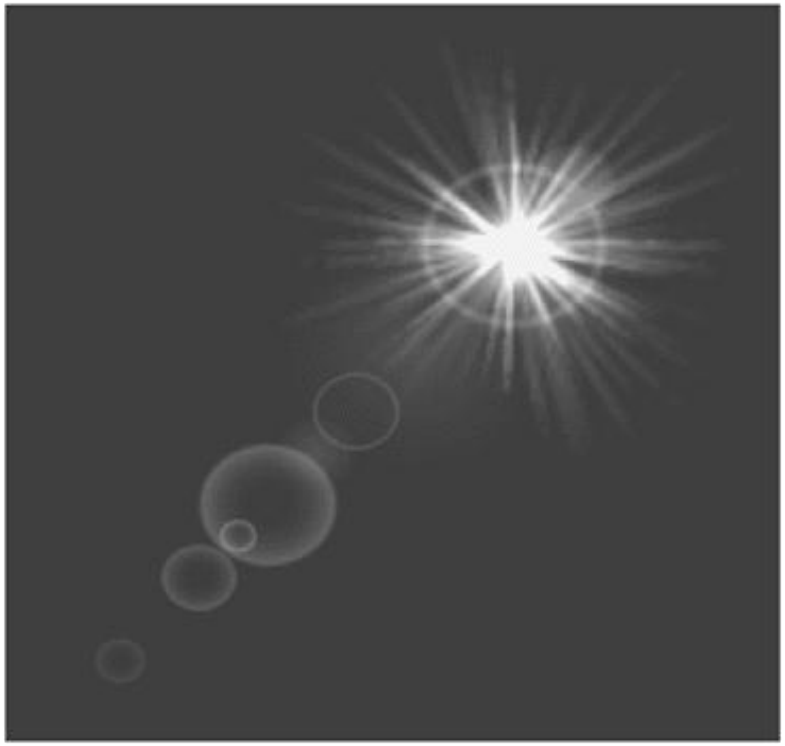
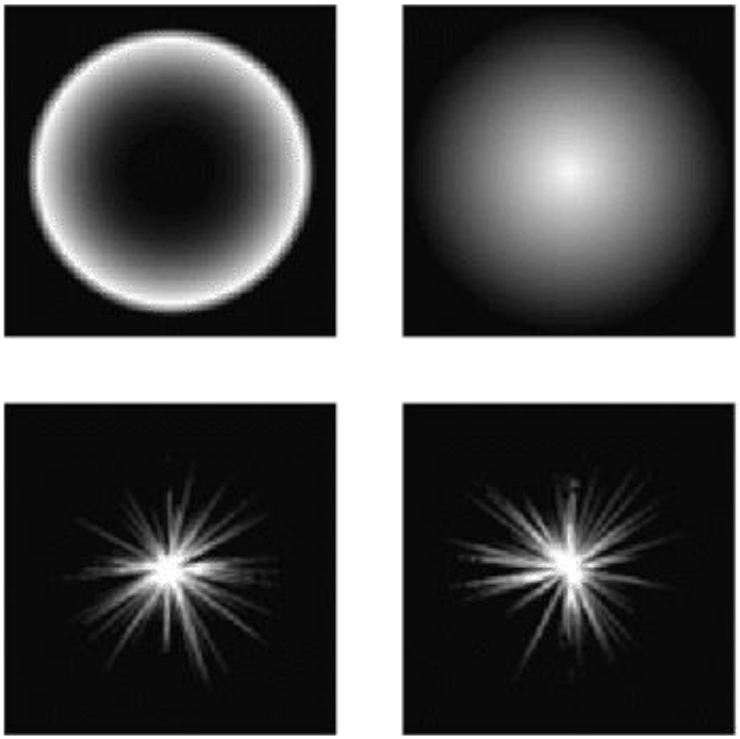
- Billboarding : Imposters
  - An imposter billboard is a billboard that is created on the fly by rendering a complex object from the current viewpoint into an image texture
  - This is then mapped onto the billboard
  - The rendering is therefore proportional to the number of pixels the imposter covers on the screen, instead of the number of vertices or the depth complexity of the object
  - We have covered imposters before when discussing "Level of Detail"

Image-Based Rendering

- Billboarding : Lens Flare
  - Lens flare is a phenomenon that is caused by the lens of the eye or camera when directed at bright light (such as the sun)
  - It consists of a halo and a ciliary corona
  - The halo appears because the lens material refracts light of different wavelengths by different amounts, as we obtain using a prism
  - The halo looks like a ring around the light, with its outside edge tinged red and inside with violet

Image-Based Rendering

- Billboarding : Lens Flare
  - The ciliary corona comes from the density fluctuations in the lens, and appears as rays radiating from a point, which may extend beyond the halo
  - We can produce lens flare by using a set of textures for the glare effects; each texture is applied to a square that is made to face the viewer
  - The texture is treated as an alpha map that determines how much of the square to blend into the scene

Image-Based Rendering

- Billboarding : Lens Flare
  - **Example:** the textures and the final composite image